Course Code / Name: SA 340 Soil and Water Management

Background and Rationale

- The aim of lectures in Soil Science is to give the student basic understanding of the genesis of soil, properties of the soil that affect its use, how to manage different soils for sustainable use, how soil serves as a life support system for plants and animals. The course is both qualitative and quantitative.

Learning Outcomes

At the completion of the course students should be able to:

- Identify different soils in the field
- Identify soils that may have high production potential
- Advise land users how much fertilizer they need to apply to their soil
- Advise those who are in irrigation how much water they need to apply and the interval of irrigation they need to follow
- Advise land users on soil and water conservation methods
- master fundamental laws and equations in hydrology and their application in typical water management situations
- identify the most important procedures of water treatment and purification and appreciate their importance with regard to possible toxicological impact on human population

Recommended References:

1. **Hillel, D. (1998). Environmental Soil Physics. Academic Press, Inc. San Diego, California.**
2. **Hillel, D. 2004. Introduction to environmental Soil physics. Elsevier.**
3. **Carter, M.R., ed. 1993. Soil sampling and methods of analysis. Canadian Society of Soil Science. 823 pp.**

Brady, N. C., and Weil, R. R. (2002). "The nature and properties of soils." 13th Ed. Prentice Hall, New Jersey